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Power Dropoff Analysis: The On/Off-Drug Variable Chain

Technical analysis of the mechanism by which carryover effects reduce statistical power in N-of-1 trial designs

1. Overview

The power to detect a biomarker-treatment interaction in N-of-1 and crossover trial designs depends entirely on the contrast between on-drug and off-drug observations. Carryover effects erode this contrast by allowing residual drug effect to persist into off-drug periods. This document traces the exact variable chain – from trial design specification through derived quantities to the analysis model interaction term – that mediates the power dropoff.

Five derived variables form this chain:

1. `ondrug / treatment` – binary path assignment
2. `tod / weeks_on_drug` – cumulative time on drug
3. `tsd / time_since_discontinuation` – time off drug
4. `Dbc / effective_drug_weeks` – continuous treatment indicator (the analysis predictor)
5. The interaction term (`bm * treatment_indicator`) – the coefficient whose significance defines power

Each variable transforms the preceding one. Carryover enters at step 3 and propagates through steps 4 and 5 to reduce the variance of the interaction term, inflating its standard error and thus reducing power.

2. Variable 1: Path Assignment (`ondrug / treatment`)

The binary indicator of whether a participant receives active drug at each measurement timepoint. This is the only variable that differs across paths within a design.

2.1 orig implementation

Paths are defined as explicit binary vectors passed to `buildtrialdesign()`:

```
# N-of-1 (Hybrid) design, 8 timepoints
# Weeks: 4, 8, 9, 10, 11, 12, 16, 20
tdNof11 <- buildtrialdesign(
  ondrug = list(
    pathA = c(1,1,1,1,0,0,1,0),
    pathB = c(1,1,1,1,0,0,0,1),
    pathC = c(1,1,1,0,0,0,1,0),
    pathD = c(1,1,1,0,0,0,0,1)
  )
)
```

Each path vector has length equal to the number of timepoints. Participants are randomly assigned to one path. The function `buildtrialdesign()` then derives `tod`, `tsd`, and `tpb` from these vectors.

2.2 main implementation

Paths are constructed procedurally in design-specific functions. The Hybrid design:

```
create_hybrid_design <- function(n_participants,
                                measurement_weeks) {
  path_assignment <- sample(rep(1:4, length.out = n_participants))

  expand_grid(
    participant_id = 1:n_participants,
    week = measurement_weeks
  ) %>%
  mutate(
    path = path_assignment[participant_id],
    treatment = case_when(
      week %in% c(4, 8) ~ 1,           # OL phase: all on drug
      week == 9 ~ 1,                 # BD phase start
      week == 10 & path %in% c(1,2) ~ 1,
      week == 10 & path %in% c(3,4) ~ 0,
      week %in% c(11, 12) ~ 0,       # All off drug
      week == 16 & path %in% c(1,3) ~ 1,
      week == 16 & path %in% c(2,4) ~ 0,
      week == 20 & path %in% c(1,3) ~ 0,
      week == 20 & path %in% c(2,4) ~ 1,
      TRUE ~ NA_real_
    ),
    expectancy = if_else(week %in% c(4, 8), 1.0, 0.5)
  )
}
```

2.3 simple implementation

Uses a switch-based `randomize_assignment()` with block randomization per design segment:

```
randomize_assignment <- function(design) {
  on_drug <- design$on_drug_fixed
  if (design$name == "C0") {
    if (runif(1) < 0.5)
      on_drug <- c(rep(TRUE, 4), rep(FALSE, 4))
    else
      on_drug <- c(rep(FALSE, 4), rep(TRUE, 4))
  }
  # ... similar for N-of-1 and OL+BDC
  on_drug
}
```

2.4 Path structure comparison

The treatment indicator across the 8 timepoints for each design's paths:

HYBRID (N-of-1) -- 4 paths

```

Week:      4   8   9  10  11  12  16  20
Path A:    1   1   1   1   0   0   1   0
Path B:    1   1   1   1   0   0   0   1
Path C:    1   1   1   0   0   0   1   0
Path D:    1   1   1   0   0   0   0   1
-----  OL phase  BD phase    CO phase
          (all on) (staggered) (crossed)

```

OL+BDC -- 2 paths (orig) / 4 paths (main)

```

Week:      4   8  12  16  17  18  19  20
Path A:    1   1   1   1   1   1   0   0
Path B:    1   1   1   1   1   0   0   0
-----  OL phase (all on)    BD (off)

```

main extends to 4 paths with finer staggering at
weeks 17-18

CROSSOVER -- 2 paths

```

Week:     2.5   5   7.5  10  12.5  15  17.5  20
Path A:    1     1   1     1   0     0   0     0
Path B:    0     0   0     0   1     1   1     1

```

OPEN-LABEL -- 1 path

```

Week:     2.5   5   7.5  10  12.5  15  17.5  20
Path A:    1     1   1     1   1     1   1     1

```

The number of off-drug timepoints per design determines the upper bound on treatment contrast:

Design	On-drug TPs	Off-drug TPs	Treatment variance
OL	8	0	0 (no contrast)
CO	4	4	Maximum (balanced)
OL+BDC	~6	~2	Low (late only)
Hybrid	~5	~3	Moderate (spread)

3. Variable 2: Cumulative Time on Drug (`tod` / `weeks_on_drug`)

This is the cumulative sum of treatment-weighted time intervals. It increases while on drug and plateaus (freezes) when off drug. It feeds the Gompertz growth curve to determine the biological response (BR) mean at each timepoint.

3.1 orig implementation

In `buildtrialdesign()` (lines 90-96):

```

od_intervals <- t_wk * od      # interval duration * ondrug
tod <- od_intervals
for (iTP in 2:length(timepoints)) {
  if (od_intervals[iTP] > 0) {
    tod[iTP] <- tod[iTP] + tod[iTP-1] # on-drug: accumulate
  }
  # off-drug: tod stays at previous accumulated value
}

```

3.2 main implementation

In `simulation_clustered.R` (line 655):

```
weeks_on_drug = cumsum(treatment)
```

Note: the main uses a counting-based cumulative sum (number of on-drug timepoints) rather than a time-weighted sum. For evenly spaced designs this is proportional; for clustered designs, the distinction matters.

3.3 Worked example: Hybrid Path A

Week:	4	8	9	10	11	12	16	20
treatment:	1	1	1	1	0	0	1	0
weeks_on_drug:	1	2	3	4	4	4	5	5
	↑ accumulating ↑			↑frozen↑		↑+1	↑frozen	

The BR mean is computed from `tod`:

```
BR_mean = modgompertz(tod, max, disp, rate)
```

Off-drug, `tod` freezes at 4, so the Gompertz-derived BR mean plateaus. Without carryover, the BR mean drops to zero immediately when treatment stops. With carryover, the plateau value decays exponentially (see Variable 3).

4. Variable 3: Time Since Discontinuation (`tsd`)

This is the elapsed time since the participant was last on drug. It is the direct input to the carryover decay function.

4.1 orig implementation

In `buildtrialdesign()` (lines 92-105):

```

tsd <- t_wk * (od != 1)      # interval * off-drug
for (iTP in 2:length(timepoints)) {
  if (od_intervals[iTP] == 0) { # off-drug
    tsd[iTP] <- tsd[iTP] + tsd[iTP-1] # accumulate
  }
  # on-drug: tsd resets (stays at interval contribution)
}

```

```
everondrug <- (cumulative(od) > 0)
tsd <- everondrug * tsd           # zero out pre-drug periods
```

4.2 main implementation

In simulation_clustered.R (lines 658-665):

```
just_discontinued = treatment == 0 &
                      lag(treatment, default = 0) == 1,
discontinuation_week = if_else(
  just_discontinued, week, NA_real_),
discontinuation_week = zoo::na.locf(
  discontinuation_week, na.rm = FALSE),
tsd = if_else(
  treatment == 0 & !is.na(discontinuation_week),
  week - discontinuation_week,
  0
)
```

The main uses absolute week differences rather than accumulated intervals. This is more precise for irregularly spaced timepoints.

4.3 simple implementation

In generate_participant():

```
if (on_drug[w]) {
  last_drug_week <- w
  carryover_factors[w] <- 0
} else if (last_drug_week > 0) {
  weeks_since <- w - last_drug_week
  carryover_factors[w] <- calc_carryover(
    weeks_since, carryover_half-life)
}
```

4.4 Worked example: Hybrid Path A

Week:	4	8	9	10	11	12	16	20
treatment:	1	1	1	1	0	0	1	0
tsd:	0	0	0	0	1	2	0	4

Carryover at $t_{\frac{1}{2}} = 0.1$ weeks (Hendrickson):

```
week 11: (1/2)^(1/0.1) = (1/2)^10 = 0.001 (0.1%)
week 12: (1/2)^(2/0.1) = (1/2)^20 = 0.000001 (~0%)
week 20: (1/2)^(4/0.1) = (1/2)^40 ~ 0 (~0%)
```

Carryover at $t_{\frac{1}{2}} = 0.2$ weeks (Hendrickson):

```
week 11: (1/2)^(1/0.2) = (1/2)^5 = 0.031 (3.1%)
week 12: (1/2)^(2/0.2) = (1/2)^10 = 0.001 (0.1%)
week 20: (1/2)^(4/0.2) = (1/2)^20 ~ 0 (~0%)
```

Carryover at $t_{1/2} = 1$ week (simple's values):

```
week 11: (1/2)^(1/1) = 0.5 (50%)
week 12: (1/2)^(2/1) = 0.25 (25%)
week 20: (1/2)^(4/1) = 0.063 (6.3%)
```

This example reveals why the half-life parameterization matters so much. At Hendrickson's values (0.1-0.2 weeks), carryover is negligible after one week of discontinuation. At the simple's values (0.5-2 weeks), substantial drug effect persists for weeks, heavily contaminating the off-drug contrast.

4.5 Scale factor complication

The orig and main apply a scale factor (default 2) inside the carryover formula:

```
# orig (generateData.R, line 96):
brmeans[p] <- brmeans[p] +
  brmeans[p-1] * (1/2)^(scalefactor * tsd / tlhalf)

# main (pm_functions.R, line 41):
previous_effect * (1/2)^(scale_factor * tsd / halflife)
```

With `scale_factor = 2`, the effective half-life is halved. A nominal `t_half = 0.2` weeks becomes an effective `t_half = 0.1` weeks. The simple omits this scale factor, so its nominal values are also the effective values.

Effective half-lives across codebases:

Codebase	Nominal $t_{1/2}$ (wk)	Scale factor	Effective $t_{1/2}$ (wk)
orig	0.1, 0.2	2	0.05, 0.10
main*	0.1, 0.2	1**	0.10, 0.20
simple	0.5, 1, 2	1	0.50, 1.00, 2.00

*main's `simulation_clustered.R` uses its own `calculate_carryover()` without scale factor; `pm_functions.R` retains `scale_factor = 2` for the shared function.

This means the main's clustered simulation and the orig produce different carryover decay rates even for the same nominal half-life value.

5. Variable 4: Continuous Treatment Indicator

This is the variable that actually enters the analysis model as a predictor. It transforms the binary treatment state into a continuous measure of drug exposure that accounts for (or ignores) carryover.

5.1 orig: Dbc

In `lme_analysis.R` (lines 107-109):

```
data.m2[Db == TRUE, Dbc := 1]
data.m2[Db == FALSE,
```

```

Dbc := (1/2)^(op$carryover_scalefactor * tsd /
            op$carryover_t1half)]

```

Dbc is bounded [0, 1]: - On drug: 1 (full exposure) - Off drug: exponentially decaying from 1 toward 0

5.2 main: effective_drug_weeks

In simulation_clustered.R (lines 697-701):

```

effective_drug_weeks = case_when(
  treatment == 1 ~ weeks_on_drug,
  treatment == 0 & !is.na(weeks_at_discont) ~
    weeks_at_discont * carryover_effect,
  TRUE ~ 0
)

```

effective_drug_weeks is bounded [0, max(weeks_on_drug)]: - On drug: cumulative count (1, 2, 3, ...) - Off drug: accumulated count at discontinuation * exponential decay

The misspecified alternative (model_carryover = FALSE) uses weeks_on_drug directly, which plateaus off-drug rather than decaying.

5.3 simple: carryover adjustment (optional)

In analyze_trial():

```

# With adjustment:
adjusted_response = response - carryover * br_rate

```

```

# Without adjustment:
# raw response used directly

```

Rather than modifying the predictor, the simple subtracts the estimated carryover effect from the response variable before computing phase means.

5.4 Worked example: Hybrid Path A

Week:	4	8	9	10	11	12	16	20
treatment:	1	1	1	1	0	0	1	0

orig Dbc ($t_{1/2}=0.2$, sf=2):	1	1	1	1	.001	~0	1	~0
-----------------------------------	---	---	---	---	------	----	---	----

main effective_drug_weeks ($t_{1/2}=0.2$):								
weeks_on_drug:	1	2	3	4	4	4	5	5
carryover_effect:	1	1	1	1	.031	.001	1	~0
effective:	1	2	3	4	.124	.004	5	~0

main weeks_on_drug (no carryover model):	1	2	3	4	4	4	5	5
--	---	---	---	---	---	---	---	---

Note the difference between the two main scenarios: - `effective_drug_weeks` drops sharply off-drug (good contrast with on-drug) - `weeks_on_drug` stays at 4 off-drug (poor contrast: the model sees accumulated exposure = 4 both on and off drug)

6. Variable 5: The Interaction Term

The coefficient whose p-value defines statistical power.

6.1 Analysis model formulas

orig (`lme_analysis.R`):

When treatment varies across participants (CO, Hybrid, OL+BDC):

```
Sx ~ bm + t + Dbc + bm * Dbc + (1 | ptID)
```

Interaction of interest: `bm:Dbc`

When treatment is constant (OL):

```
Sx ~ bm + t + bm * t + (1 | ptID)
```

Interaction of interest: `bm:t`

main (`simulation_clustered.R`):

With carryover modeled:

```
response ~ effective_drug_weeks * bm_centered + week
          + (1 | participant_id)
```

or with `nlme::lme`:

```
response ~ effective_drug_weeks * bm_centered + week,
  random = ~ 1 | participant_id,
  correlation = corCAR1(form = ~ week | participant_id)
```

Interaction of interest: `effective_drug_weeks:bm_centered`

Without carryover modeled:

```
response ~ weeks_on_drug * bm_centered + week
          + (1 | participant_id)
```

Interaction of interest: `weeks_on_drug:bm_centered`

simple (`analyze_trial()`):

For CO and N-of-1:

```
drug_effect ~ biomarker
```

where `drug_effect` = `mean(on_drug) - mean(off_drug)` per participant.

For OL+BDC:

```
(blinded - open) ~ biomarker * group
```

Interaction of interest: biomarker:groupplacebo

6.2 How carryover degrades the interaction

The interaction coefficient estimates the degree to which the biomarker predicts differential response to treatment. Its standard error is inversely proportional to the variance of the treatment indicator:

$$SE(\text{beta_interaction}) \sim 1 / \sqrt{\text{Var}(\text{treatment_indicator})}$$

Carryover reduces $\text{Var}(\text{treatment_indicator})$ by making off-drug observations resemble on-drug observations:

Without carryover:

On-drug predictor values: [1, 2, 3, 4, 5]
Off-drug predictor values: [0, 0, 0]
Contrast: large

With carryover ($t_{1/2} = 1$ week):

On-drug predictor values: [1, 2, 3, 4, 5]
Off-drug predictor values: [2.0, 1.0, 0.3]
Contrast: reduced

With carryover ($t_{1/2} = 2$ weeks):

On-drug predictor values: [1, 2, 3, 4, 5]
Off-drug predictor values: [2.8, 2.0, 1.2]
Contrast: severely reduced

When the treatment indicator has low variance, the regression cannot distinguish the biomarker-moderated drug effect from noise, and power drops.

7. Design-Specific Vulnerability

7.1 Open-Label (OL)

No off-drug timepoints exist. The interaction uses $\text{bm} * \text{t}$ (biomarker x elapsed time) rather than $\text{bm} * \text{treatment_indicator}$. Carryover is irrelevant because there is no discontinuation.

Power depends on: biomarker-time interaction strength. Carryover sensitivity: **none**.

7.2 Traditional Crossover (CO)

Four on-drug and four off-drug timepoints provide maximum treatment contrast. However, the entire second period is vulnerable to carryover from the first period. For Path A (drug first), all off-drug measurements (weeks 12.5-20) carry residual drug effect.

Path A (drug -> placebo):

Week:	2.5	5	7.5	10	12.5	15	17.5	20
Tx:	1	1	1	1	0	0	0	0
tsd:	0	0	0	0	2.5	5.0	7.5	10
CO at $t_{1/2}=0.2$:	--	--	--	--	.0001	~0	~0	~0
CO at $t_{1/2}=2$:	--	--	--	--	.42	.18	.07	.03

At short half-lives (Hendrickson), carryover is negligible by the first off-drug measurement. At long half-lives (simple), substantial contamination persists.

Power depends on: on/off contrast in second half. Carryover sensitivity: **high** (long tsd values help with Hendrickson values; problematic with longer $t_{1/2}$).

7.3 Open-Label + Blinded Discontinuation (OL+BDC)

Long open-label phase (weeks 4-16 on drug) followed by short blinded discontinuation (weeks 17-20, staggered off-drug). Only 2-4 off-drug timepoints exist, all at the end of the trial.

OL+BDC Path A (orig):

Week:	4	8	12	16	17	18	19	20
Tx:	1	1	1	1	1	1	0	0
tsd:	0	0	0	0	0	0	1	2

The tsd values are short (1-2 weeks from last on-drug timepoint to off-drug measurements), which means even modest carryover half-lives can produce substantial contamination:

C0 at $t_{1/2}=0.1$: week 19: 0.001, week 20: ~0
 C0 at $t_{1/2}=0.2$: week 19: 0.031, week 20: 0.001
 C0 at $t_{1/2}=1.0$: week 19: 0.500, week 20: 0.250

Power depends on: late on/off contrast with few off-drug timepoints. Carryover sensitivity: **moderate to high** (few off-drug timepoints; short tsd means even moderate $t_{1/2}$ contaminates).

7.4 Hybrid (N-of-1)

Combines open-label, blinded discontinuation, and crossover phases. Four paths create varied discontinuation patterns:

Hybrid Path A:

Week:	4	8	9	10	11	12	16	20
Tx:	1	1	1	1	0	0	1	0
tsd:	0	0	0	0	1	2	0	4

Hybrid Path D:

Week:	4	8	9	10	11	12	16	20
Tx:	1	1	1	0	0	0	0	1
tsd:	0	0	0	0	1	2	3	0

Multiple discontinuation events create multiple contrast windows with different tsd values, providing the analysis model with richer information about the decay trajectory.

Power depends on: multiple contrast windows across phases. Carryover sensitivity: **moderate** (varied tsd values mean some contrasts are clean even when others are contaminated).

8. Summary Table

+-----+	+-----+	+-----+	+-----+	+-----+	+-----+
Variable	orig	main	simple	Role	

treatment	ondrug	treatment	on_drug	Binary	
(V1)	(list of	(case_	(block	path	
	vectors)	when)	random)	assignment	
cum. drug	tod	weeks_on	(not	Gompertz	
time (V2)	(accum	_drug	cumul.)	input for	
	loop)	(cumsum)		BR mean	
time off	tsd	tsd	weeks_	Carryover	
drug (V3)	(accum	(week -	since	decay	
	loop)	discount)		input	
continuous	Dbc	effective	carryover	Analysis	
treatment	[0,1]	_drug_wks	_factors	model	
(V4)		[0,max]	[0,1]	predictor	
interact.	bm:Dbc	eff_drug	biomarker	Coeff	
term (V5)	or bm:t	:bm_cent	(on diff)	tested	

9. Variable Equivalence: orig vs. main

This section traces each derived variable through both codebases for every design, using the exact inputs from the orig vignette and the main's `simulation_clustered.R`. The goal is to determine whether the two produce identical treatment-related variables.

9.1 Hybrid (N-of-1) Design

Inputs (orig vignette, lines 48-61):

```
timepoints = c(4, 8, 9, 10, 11, 12, 16, 20)
ondrug = list(
  pathA = c(1,1,1,1,0,0,1,0),
  pathB = c(1,1,1,1,0,0,0,1),
  pathC = c(1,1,1,0,0,0,1,0),
  pathD = c(1,1,1,0,0,0,0,1)
)
expectancies = c(1, 1, .5, .5, .5, .5, .5, .5)
```

Inputs (main, lines 366, 394-417):

```
measurement_weeks_hybrid = c(4, 8, 9, 10, 11, 12, 16, 20)
treatment = case_when(
  week %in% c(4, 8) ~ 1,
  week == 9 ~ 1,
  week == 10 & path %in% c(1, 2) ~ 1,
  week == 10 & path %in% c(3, 4) ~ 0,
```

```

week %in% c(11, 12) ~ 0,
week == 16 & path %in% c(1, 3) ~ 1,
week == 16 & path %in% c(2, 4) ~ 0,
week == 20 & path %in% c(1, 3) ~ 0,
week == 20 & path %in% c(2, 4) ~ 1
)

```

9.1.1 treatment (V1): EQUAL

Week:	4	8	9	10	11	12	16	20	
orig A:	1	1	1	1	0	0	1	0	
main 1:	1	1	1	1	0	0	1	0	✓

orig B:	1	1	1	1	0	0	0	1	
main 2:	1	1	1	1	0	0	0	1	✓

orig C:	1	1	1	0	0	0	1	0	
main 3:	1	1	1	0	0	0	1	0	✓

orig D:	1	1	1	0	0	0	0	1	
main 4:	1	1	1	0	0	0	0	1	✓

The binary treatment vectors are identical across all four paths.

9.1.2 tod / weeks_on_drug (V2): NOT EQUAL The orig computes *tod* as cumulative *time-weighted* exposure (in weeks). The main computes *weeks_on_drug* as a simple *count* of on-drug timepoints via `cumsum(treatment)`.

orig buildtrialdesign() logic:

```

t_wk = c(4, 4, 1, 1, 1, 1, 4, 4) # interval durations
od_intervals = t_wk * od          # weighted by ondrug

```

For Path A (`od = c(1,1,1,1,0,0,1,0)`):

```
od_intervals = c(4, 4, 1, 1, 0, 0, 4, 0)
```

Accumulation loop: `tod[i] = tod[i] + tod[i-1]` when `od_intervals[i] > 0`:

```

i=1: tod = 4
i=2: od_intervals[2]=4 > 0: tod = 4 + 4 = 8
i=3: od_intervals[3]=1 > 0: tod = 1 + 8 = 9
i=4: od_intervals[4]=1 > 0: tod = 1 + 9 = 10
i=5: od_intervals[5]=0:    tod = 0 (no accumulation)
i=6: od_intervals[6]=0:    tod = 0
i=7: od_intervals[7]=4 > 0: tod = 4 + 0 = 4
    ^^ RESETS! Does NOT carry forward from i=4.
i=8: od_intervals[8]=0:    tod = 0

```

Wait – that cannot be right. Let me re-read the loop more carefully. The accumulation is:

```

tod <- od_intervals           # initialize
for (iTP in 2:length) {
  if (od_intervals[iTP] > 0) {
    tod[iTP] <- tod[iTP] + tod[iTP-1]
  }
}

```

So `tod[iTP]` carries forward from `tod[iTP-1]`, not from the last on-drug value. At $i=7$, `tod[iTP-1]` = `tod[6]` = 0, so `tod[7]` = $4 + 0 = 4$.

orig tod for Path A:

Week:	4	8	9	10	11	12	16	20
tod:	4	8	9	10	0	0	4	0

main weeks_on_drug for Path 1:

```

weeks_on_drug = cumsum(treatment)
treatment:      1  1  1  1  0  0  1  0
cumsum:         1  2  3  4  4  4  5  5

```

Week:	4	8	9	10	11	12	16	20
wod:	1	2	3	4	4	4	5	5

These are NOT equal. They differ in two fundamental ways:

1. **Units.** The orig's `tod` is in *weeks* of drug exposure (weighted by interval duration). The main's `weeks_on_drug` is a *count* of on-drug measurement occasions.
2. **Continuity across gaps.** The orig's `tod` resets after an off-drug gap because it only carries forward from `tod[iTP-1]`, which is 0 during off-drug periods. After resuming drug at week 16, `tod` = 4 (the 4-week interval), not $10 + 4 = 14$. The main's `cumsum` never resets – it monotonically increases, reaching 5 at week 16.

Consequence for BR mean computation:

In the orig, BR mean is computed via `modgompertz(tod, ...)`. After an off-drug gap, `tod` restarts from the current interval duration. This means the biological response curve *restarts* from a low value after re-initiation, as though the previous drug exposure was partially forgotten.

In the main, BR mean is computed as `weeks_on_drug * effective_BR_rate`. Since `weeks_on_drug` monotonically increases, the BR mean after re-initiation continues from the accumulated count (5, not restarting from 1). The growth trajectory does not reset.

This is a **substantive methodological difference** in how drug re-initiation is modeled. The orig treats each drug episode as partially independent (Gompertz restarts from a reduced `tod`). The main treats total exposure as cumulative regardless of gaps.

9.1.3 tsd (V3): NOT EQUAL orig logic:

```

tsd <- t_wk * (od != 1)      # interval * off-drug
for (iTP in 2:length) {
  if (od_intervals[iTP] == 0) { # off-drug
    tsd[iTP] <- tsd[iTP] + tsd[iTP-1]
  }
}

```

```

}
}
everondrug <- (cumulative(od) > 0)
tsd <- everondrug * tsd

For Path A:

t_wk = c(4, 4, 1, 1, 1, 1, 4, 4)
od    = c(1, 1, 1, 1, 0, 0, 1, 0)

tsd_init = t_wk * (od != 1) = c(0, 0, 0, 0, 1, 1, 0, 4)

```

```

Accumulation (only when od_intervals[i] == 0):
i=2: od_int[2]=4 > 0, skip
i=3: od_int[3]=1 > 0, skip
i=4: od_int[4]=1 > 0, skip
i=5: od_int[5]=0: tsd[5] = 1 + tsd[4] = 1 + 0 = 1
i=6: od_int[6]=0: tsd[6] = 1 + tsd[5] = 1 + 1 = 2
i=7: od_int[7]=4 > 0, skip
i=8: od_int[8]=0: tsd[8] = 4 + tsd[7] = 4 + 0 = 4

```

```

everondrug = cumulative(c(1,1,1,1,0,0,1,0))
            = c(1, 2, 3, 4, 4, 4, 5, 5) > 0
            = c(T, T, T, T, T, T, T, T)
tsd = everondrug * tsd (all TRUE, no effect)

```

orig tsd for Path A:

Week:	4	8	9	10	11	12	16	20
tsd:	0	0	0	0	1	2	0	4

main logic (lines 658-665):

```

just_discontinued = treatment == 0 &
                    lag(treatment, default = 0) == 1
discontinuation_week = if_else(just_discontinued,
                                week, NA_real_)
discontinuation_week = zoo::na.locf(discontinuation_week,
                                    na.rm = FALSE)

tsd = if_else(
  treatment == 0 & !is.na(discontinuation_week),
  week - discontinuation_week, 0)

```

For Path 1:

Week:	4	8	9	10	11	12	16	20
treatment:	1	1	1	1	0	0	1	0
lag(treatment):	0	1	1	1	1	0	0	1
just_discont:	F	F	F	F	T	F	F	T

```

discontinuation_week before locf:
      NA  NA  NA  NA  11  NA  NA  20

```

discontinuation_week after locf:

NA NA NA NA 11 11 11 20

tsd:

wk 4: treatment=1 -> 0
wk 8: treatment=1 -> 0
wk 9: treatment=1 -> 0
wk 10: treatment=1 -> 0
wk 11: treatment=0, discount_wk=11: 11-11 = 0
wk 12: treatment=0, discount_wk=11: 12-11 = 1
wk 16: treatment=1 -> 0
wk 20: treatment=0, discount_wk=20: 20-20 = 0

main tsd for Path 1:

Week:	4	8	9	10	11	12	16	20
tsd:	0	0	0	0	0	1	0	0

orig vs. main tsd for Path A/1:

Week:	4	8	9	10	11	12	16	20
orig:	0	0	0	0	1	2	0	4
main:	0	0	0	0	0	1	0	0
					^^^	^^^		^^^

These are NOT equal. They differ at weeks 11, 12, and 20:

- **Week 11:** orig = 1, main = 0. The orig counts the interval from week 10 to week 11 (1 week) as time since drug. The main computes $\text{week} - \text{discontinuation_week} = 11 - 11 = 0$ because the discontinuation happens *at* week 11 (the first off-drug measurement), so tsd = 0 at the moment of discontinuation.
- **Week 12:** orig = 2, main = 1. Difference of 1 week, consistent with the offset at week 11.
- **Week 20:** orig = 4, main = 0. The orig accumulates 4 weeks of off-drug time (the interval from week 16 to week 20). The main sets $\text{discontinuation_week} = 20$ (week 20 is the first off-drug timepoint after resuming at week 16), so $\text{tsd} = 20 - 20 = 0$.

Root cause: The two codebases define “time since discontinuation” differently:

- **orig:** tsd at a given timepoint is the cumulative off-drug *interval duration* leading up to and including that timepoint. It measures “how much off-drug time has elapsed since the last on-drug interval ended.”
- **main:** tsd at a given timepoint is $\text{week} - \text{discontinuation_week}$, where $\text{discontinuation_week}$ is the *week number of the first off-drug observation*. At the first off-drug timepoint itself, $\text{tsd} = 0$.

The orig’s convention is that tsd represents the time *already elapsed* off drug by the time a measurement is taken. The main’s convention is that tsd represents the time elapsed since the *transition point*.

This produces a systematic offset: **the orig's tsd is approximately one measurement interval larger than the main's at every off-drug timepoint.** For the Hybrid design's BD phase (1-week intervals), the offset is 1 week. For the CO phase (4-week interval at week 20), the offset is 4 weeks.

9.1.4 Db (derived on-drug indicator): NOT EQUAL The orig derives Db in `lme_analysis.R` (line 102) as:

```
Db = (tod > 0)
```

For Path A:

Week:	4	8	9	10	11	12	16	20
tod:	4	8	9	10	0	0	4	0
Db:	T	T	T	T	F	F	T	F

The main uses `treatment` directly (which equals `ondrug`):

Week:	4	8	9	10	11	12	16	20
tx:	1	1	1	1	0	0	1	0

These happen to be equal for Path A because `tod > 0` coincides with `ondrug == 1` (`tod` is zero exactly when `ondrug` is zero). **Equal for this design.**

9.1.5 Carryover effect: NOT EQUAL The carryover decay applied to the BR mean uses the `tsd` values computed above. Since `tsd` differs, the carryover effect differs.

orig carryover (generateData.R, line 96):

```
brmeans[p] <- brmeans[p] +
  brmeans[p-1] * (1/2)^(scalefactor * tsd[p] / t1half)
```

With `scalefactor = 2`, `t1half = 0.2`:

Week 11: `tsd=1`: $(1/2)^{(2*1/0.2)} = (1/2)^{10} = 0.00098$
 Week 12: `tsd=2`: $(1/2)^{(2*2/0.2)} = (1/2)^{20} \approx 0$
 Week 20: `tsd=4`: $(1/2)^{(2*4/0.2)} = (1/2)^{40} \approx 0$

main carryover (simulation_clustered.R, line 679):

```
carryover_effect = (1/2)^(tsd / carryover_t_half)
```

With `t1half = 0.2`:

Week 11: `tsd=0`: `carryover_effect = 1` (on-drug formula applies since `treatment=0` but `tsd=0` means the case_when hits: $(1/2)^{(0/0.2)} = 1$)
 But wait -- `treatment=0`, so this goes to the off-drug branch. $(1/2)^{(0/0.2)} = (1/2)^0 = 1$
 Week 12: `tsd=1`: $(1/2)^{(1/0.2)} = (1/2)^5 = 0.03125$
 Week 20: `tsd=0`: $(1/2)^{(0/0.2)} = 1$

Comparison at $t^{1/2} = 0.2$ weeks:

Week:	11	12	20
orig C0:	0.00098	~0	~0
main C0:	1.000	0.031	1.000
	~~~~	~~~~	~~~~

**These are dramatically different.** The main computes a carryover_effect of 1.0 (full retention) at weeks 11 and 20 because  $tsd = 0$  at those timepoints. The orig computes near-zero retention because its  $tsd$  is 1 and 4 weeks respectively.

The consequence is that the main's BR mean at week 11 (first off-drug measurement) retains the full accumulated drug effect, while the orig's BR mean decays it almost completely. At week 20 (off-drug after re-initiation), the main retains the full effect while the orig decays it to zero.

This is a **critical divergence** that directly affects power calculations. The main's model allows much more residual drug effect during off-drug periods than the orig's model, particularly at the first off-drug measurement after each discontinuation.

### 9.1.6 expectancy: EQUAL

Week:	4	8	9	10	11	12	16	20
orig:	1	1	.5	.5	.5	.5	.5	.5
main:	1	1	.5	.5	.5	.5	.5	.5 ✓

## 9.2 OL+BDC Design

**Inputs (orig vignette, lines 75-85):**

```
timepoints = c(4, 8, 12, 16, 17, 18, 19, 20)
ondrug = list(
  pathA = c(1,1,1,1,1,1,0,0),
  pathB = c(1,1,1,1,1,0,0,0)
)
expectancies = c(1, 1, 1, 1, .5, .5, .5, .5)
```

**Inputs (main, lines 373-392):**

```
measurement_weeks_ol_bdc = c(4, 8, 12, 16, 17, 18, 19, 20)
treatment = case_when(
  week <= 16 ~ 1,
  week == 17 ~ 1,
  week == 18 & path %in% c(1, 2) ~ 1,
  week == 18 & path %in% c(3, 4) ~ 0,
  week %in% c(19, 20) ~ 0,
  TRUE ~ 1
)
```

**9.2.1 treatment (V1): NOT EQUAL (different paths)** The orig defines **2 paths**:

Week:	4	8	12	16	17	18	19	20
orig A:	1	1	1	1	1	1	0	0
orig B:	1	1	1	1	1	0	0	0

The main defines **4 paths** with a different staggering:

```

Week:      4  8 12 16 17 18 19 20
main 1:    1  1  1  1  1  1  0  0
main 2:    1  1  1  1  1  1  0  0
main 3:    1  1  1  1  1  0  0  0
main 4:    1  1  1  1  1  0  0  0

```

Main paths 1-2 are identical to orig path A; main paths 3-4 are identical to orig path B. The main uses 4 paths (equal assignment) where the orig uses 2 paths. **The effective treatment vectors are the same two patterns, but with 4-path assignment in the main vs. 2-path in the orig.** The statistical content is equivalent – both produce a 50/50 split between the two discontinuation schedules.

However, there is a subtle difference at week 17:

```

orig A: week 17, ondrug = 1
orig B: week 17, ondrug = 1
main 1: week 17, treatment = 1 (week <= 16 fails,
      week == 17 ~ 1)
main 3: week 17, treatment = 1 (same)

```

Both agree: all paths are on drug at week 17. The first off-drug measurement is at week 18 (paths B/3-4) or week 19 (paths A/1-2).

**The 2-vs-4 path distinction is cosmetic for OL+BDC.** Both produce identical treatment patterns at 50/50 ratio.

### 9.2.2 tod / weeks_on_drug (V2): NOT EQUAL orig tod for Path A:

```

t_wk = c(4, 4, 4, 4, 1, 1, 1, 1)
od    = c(1, 1, 1, 1, 1, 1, 0, 0)

```

```

od_intervals = c(4, 4, 4, 4, 1, 1, 0, 0)

```

tod accumulation:

```

i=1: 4
i=2: 4+4 = 8
i=3: 4+8 = 12
i=4: 4+12 = 16
i=5: 1+16 = 17
i=6: 1+17 = 18
i=7: od_int=0, no accum: tod = 0
i=8: od_int=0, no accum: tod = 0

```

```

Week:    4    8   12   16   17   18   19   20
tod:     4    8   12   16   17   18    0    0

```

**main weeks_on_drug for Path 1:**

```

Week:    4    8   12   16   17   18   19   20
tx:      1    1    1    1    1    1    0    0
cumsum:  1    2    3    4    5    6    6    6

```

**Differences:** Same two issues as Hybrid: 1. Units (weeks vs. count) 2. Off-drug behavior: orig drops to 0, main plateaus at 6

### 9.2.3 tsd (V3): NOT EQUAL orig tsd for Path A:

```
tsd_init = t_wk * (od != 1) = c(0,0,0,0,0,0,1,1)
```

Accumulation:

```
i=7: od_int=0: tsd = 1 + tsd[6] = 1 + 0 = 1
```

```
i=8: od_int=0: tsd = 1 + tsd[7] = 1 + 1 = 2
```

Week:	4	8	12	16	17	18	19	20
tsd:	0	0	0	0	0	0	1	2

main tsd for Path 1:

Week:		4	8	12	16	17	18	19	20
treatment:		1	1	1	1	1	1	0	0
lag(treatment):	0	1	1	1	1	1	1	1	0
just_discont:	F	F	F	F	F	F	F	T	F
discont_week:	NA	NA	NA	NA	NA	NA	NA	19	19
tsd:		0	0	0	0	0	0	0	1

  

Week:	4	8	12	16	17	18	19	20
orig:	0	0	0	0	0	0	1	2
main:	0	0	0	0	0	0	0	1
							^^	^^

Same systematic offset: the orig's tsd is one interval ahead of the main's. At the first off-drug timepoint (week 19), orig says tsd = 1 (one week has elapsed off drug) while main says tsd = 0 (just discontinued).

## 9.3 Traditional Crossover (CO) Design

**Inputs (orig vignette, lines 87-98):**

```
timepoints = cumulative(rep(2.5, 8))
# = c(2.5, 5, 7.5, 10, 12.5, 15, 17.5, 20)
ondrug = list(
  pathA = c(1,1,1,1,0,0,0,0),
  pathB = c(0,0,0,0,1,1,1,1)
)
expectancies = rep(.5, 8)
```

**Inputs (main, lines 419-442):**

```
measurement_weeks_crossover =
  c(2.5, 5, 7.5, 10, 12.5, 15, 17.5, 20)
treatment = case_when(
  week <= 10 & path == 1 ~ 1,
  week <= 10 & path == 2 ~ 0,
  week > 10 & path == 1 ~ 0,
```

```

week > 10 & path == 2 ~ 1
)

```

### 9.3.1 treatment (V1): EQUAL

Week:	2.5	5	7.5	10	12.5	15	17.5	20
orig A:	1	1	1	1	0	0	0	0
main 1:	1	1	1	1	0	0	0	0 ✓

  

orig B:	0	0	0	0	1	1	1	1
main 2:	0	0	0	0	1	1	1	1 ✓

### 9.3.2 tod / weeks_on_drug (V2): NOT EQUAL orig tod for Path A:

```

t_wk = rep(2.5, 8)
od_intervals = c(2.5, 2.5, 2.5, 2.5, 0, 0, 0, 0)
tod: 2.5 5.0 7.5 10.0 0 0 0 0

```

main weeks_on_drug for Path 1:

```

cumsum(1,1,1,1,0,0,0,0) = 1 2 3 4 4 4 4 4

```

Different units (weeks vs. count) and different off-drug behavior (0 vs. plateau at 4).

### 9.3.3 tsd (V3): NOT EQUAL orig tsd for Path A:

```

tsd_init = c(0, 0, 0, 0, 2.5, 2.5, 2.5, 2.5)

```

Accumulation:

```

i=5: 2.5 + 0 = 2.5
i=6: 2.5 + 2.5 = 5.0
i=7: 2.5 + 5.0 = 7.5
i=8: 2.5 + 7.5 = 10.0

```

Week:	2.5	5	7.5	10	12.5	15	17.5	20
tsd:	0	0	0	0	2.5	5.0	7.5	10.0

main tsd for Path 1:

```

just_discont at week 12.5: TRUE (tx=0, lag=1)

```

discont_week:

```

NA NA NA NA 12.5 12.5 12.5 12.5

```

tsd:

```

0 0 0 0 0 2.5 5.0 7.5

```

Week:	2.5	5	7.5	10	12.5	15	17.5	20
orig:	0	0	0	0	2.5	5.0	7.5	10.0
main:	0	0	0	0	0	2.5	5.0	7.5
					^^^	^^^	^^^	^^^

Systematic offset of 2.5 weeks (one measurement interval) at every off-drug timepoint.

### 9.3.4 tsd for Path B (placebo-first): DIFFERENT ISSUE orig tsd for Path B:

```
od = c(0,0,0,0,1,1,1,1)
tsd_init = c(2.5, 2.5, 2.5, 2.5, 0, 0, 0, 0)
```

Before accumulation:

```
i=2: od_int=0: tsd = 2.5 + 2.5 = 5.0
i=3: od_int=0: tsd = 2.5 + 5.0 = 7.5
i=4: od_int=0: tsd = 2.5 + 7.5 = 10.0
i=5-8: od_int > 0: skip
```

```
everondrug = cumulative(0,0,0,0,1,1,1,1) > 0
             = c(0,0,0,0,1,2,3,4) > 0
             = c(F, F, F, F, T, T, T, T)
tsd = everondrug * tsd
     = c(0, 0, 0, 0, 0, 0, 0, 0)
```

Week:	2.5	5	7.5	10	12.5	15	17.5	20
tsd:	0	0	0	0	0	0	0	0

Path B never has off-drug time *after* being on drug, so tsd is zero everywhere. This is correct: carryover only applies after drug exposure.

#### main tsd for Path 2:

Path 2 has treatment = c(0,0,0,0,1,1,1,1). There is no discontinuation (no transition from 1 to 0), so just_discontinued is never TRUE. tsd = 0 everywhere.

Path B/2: EQUAL (both zero everywhere).

## 9.4 Open-Label (OL) Design

### Inputs (orig, lines 64-73):

```
timepoints = cumulative(rep(2.5, 8))
ondrug = list(pathA = rep(1, 8))
expectancies = rep(1, 8)
```

### Inputs (main, lines 444-457):

```
treatment = 1 # Always on drug
expectancy = 1.0
```

All variables trivially equal: treatment = 1 everywhere, tod accumulates continuously, tsd = 0 everywhere. **EQUAL**.

## 9.5 Summary of Divergences

Variable	OL	CO	Hybrid & OL+BDC
treatment	Equal	Equal	Equal

tod /	Not equal	Not equal	Not equal
weeks_on_	(units	(units +	(units +
drug	only)	off-drug	off-drug
		behavior)	behavior)
tsd	Equal	Not equal	Not equal
	(both 0)	(offset by	(offset
		1 interval)	+ reset)
Db / tx	Equal	Equal	Equal
(binary)			
carryover	Equal	Not equal	Not equal
effect	(both 0)	(offset	(offset
		propagates)	+ reset)

## 9.6 Root Causes

**Divergence 1: tod units and accumulation.** The orig accumulates actual time-on-drug in *weeks* (weighted by interval duration). The main counts on-drug *measurement occasions* via `cumsum(treatment)`. For evenly spaced designs (CO, OL) these differ by a constant factor (2.5x for CO). For irregularly spaced designs (Hybrid, OL+BDC), the relationship is non-linear because intervals range from 1 to 4 weeks.

This affects the BR mean computation. The orig feeds `tod` (in weeks) into a Gompertz curve. The main feeds `weeks_on_drug` (count) into a linear rate model (`weeks_on_drug * BR_rate`). These produce different dose-response shapes independent of the unit difference.

**Divergence 2: tod off-drug behavior.** The orig's `tod` carries forward only from the immediately preceding timepoint. During off-drug periods, `tod[i-1]` is 0 (since `tod` was not accumulated), so when drug resumes, `tod` restarts from the new interval's duration rather than from the pre-gap accumulated value. The main's `cumsum(treatment)` is monotonically non-decreasing and plateaus at the last on-drug count.

This means the orig models drug re-initiation as a partial restart (lower Gompertz input), while the main models it as continuation of cumulative exposure.

**Divergence 3: tsd measurement convention.** The orig's `tsd` at a timepoint measures the total off-drug interval duration from the last on-drug interval through the current timepoint. At the *first* off-drug measurement, this equals the interval duration (typically 1-2.5 weeks). The main's `tsd` measures elapsed time since the discontinuation week itself, so at the first off-drug measurement `tsd = 0`.

This produces a systematic offset of approximately one measurement interval. The consequence is that the main applies *more* carryover retention at the first off-drug timepoint than the orig (because  $(1/2)^0 = 1$  vs.  $(1/2)^{(sf * interval / t_half)}$ ).

**Divergence 4: Scale factor.** The orig's carryover formula includes `scalefactor = 2` by default. The main's `simulation_clustered.R` uses a direct  $(1/2)^{(tsd / t_half)}$  without scale factor. This

doubles the effective decay rate in the orig relative to the main for the same nominal half-life.

## 9.7 Combined Impact on Power Estimates

These divergences compound. For the Hybrid design at  $t_{1/2} = 0.2$  weeks, the carryover retention at the first off-drug timepoint (week 11) is:

- **orig:**  $(1/2)^{(2 * 1 / 0.2)} = (1/2)^{10} \approx 0.001$  (0.1% retention)
- **main:**  $(1/2)^{(0 / 0.2)} = (1/2)^0 = 1.0$  (100% retention)

This is a three-order-of-magnitude difference in carryover effect at the most informative contrast point. The main effectively treats the first off-drug measurement as having full drug effect, while the orig treats it as having nearly zero residual effect.

The implications for power estimation are:

1. The **main will show larger power dropoff** from carryover than the orig, because off-drug observations retain more drug effect and thus reduce the on/off contrast more severely.
2. The **orig's power estimates are less sensitive to carryover** because the first off-drug measurement already has minimal residual effect, preserving the contrast.
3. The two codebases will produce **non-comparable power curves** even for identical nominal parameters (same  $t_{1/2}$ , same design, same sample size).

Whether the orig or main convention is “correct” depends on the intended interpretation:

- If **tsd** should measure “how long has the participant been off drug by the time this measurement is taken,” the **orig's convention** is more natural – the participant has been off drug for the full interval duration by measurement time.
- If **tsd** should measure “how long since the treatment state changed,” the **main's convention** is more natural – the transition happens at the week boundary and time accumulates from there.

The Hendrickson et al. (2020) paper should be consulted to determine which convention matches the published methodology.

---

## 10. Recommendations for pmsimstats2025

The goal of pmsimstats2025 is to replicate the orig's simulation methodology as closely as possible using modern R practices, and then to diagnose why the orig shows power falling too rapidly as carryover increases. The recommendations below are ordered to achieve faithful replication first (items 1-4), then enable diagnosis of the power dropoff (items 5-6).

### 10.1 Replication: Fix tsd convention (Critical)

The main's `tsd = week - discontinuation_week` produces `tsd = 0` at the first off-drug timepoint. The orig's interval-accumulation approach produces `tsd = interval duration` (1-2.5 weeks) at the first off-drug timepoint. This is the single largest source of divergence.

Recommended fix in `simulation_clustered.R`:



```

# Replace current tsd computation (lines 658-665)
# with last-on-drug-week anchor:
last_on_drug_week = if_else(
  treatment == 1, week, NA_real_),
last_on_drug_week = zoo::na.locf(
  last_on_drug_week, na.rm = FALSE),
tsd = if_else(
  treatment == 0 & !is.na(last_on_drug_week),
  week - last_on_drug_week,
  0
)

```

This anchors `tsd` to the last *on-drug* measurement week rather than the first *off-drug* measurement week, producing values that match the orig:

Hybrid Path A:

Week:	4	8	9	10	11	12	16	20
Current:	0	0	0	0	0	1	0	0
Fixed:	0	0	0	0	1	2	0	4
Orig:	0	0	0	0	1	2	0	4 ✓

## 10.2 Replication: Time-weighted `tod` with gap reset (Critical)

Replace `cumsum(treatment)` with time-weighted accumulation that resets after off-drug gaps, matching the orig's `tod` behavior.

The orig's `tod` at timepoint *i* equals: - On drug: sum of interval durations for all consecutive on-drug timepoints ending at *i* - Off drug: 0

This means drug re-initiation restarts the exposure clock. At week 16 of Hybrid Path A, `tod` = 4 (the 4-week interval since week 12), not 14 (total lifetime exposure).

Recommended approach: compute `tod` with the same loop logic as the orig, or equivalently:

```

interval = week - lag(week, default = 0),
drug_interval = treatment * interval,
# Reset-aware cumulative sum: restarts when
# drug_interval is preceded by 0
tod = ave(drug_interval, cumsum(drug_interval == 0),
  FUN = cumsum)

```

This also changes the BR mean computation. The orig uses `modgompertz(tod, ...)` while the main uses `weeks_on_drug * BR_rate`. To replicate the orig, the main should either: - (a) Implement the modified Gompertz and feed it `tod`, matching the orig's DGP exactly, or - (b) Use `tod * BR_rate` as a linear approximation, accepting that the dose-response shape differs

Option (a) is more faithful; option (b) is simpler and may suffice if the Gompertz is approximately linear over the trial duration.

## 10.3 Replication: Restore `scalefactor = 2` (Critical)

The orig's carryover formula is:

```
brmeans[p] + brmeans[p-1] *
  (1/2)^(scalefactor * tsd / tlhalf)
```

with `scalefactor = 2`. The main's `simulation_clustered.R` omits the scale factor:

```
(1/2)^(tsd / carryover_t_half)
```

This halves the effective decay rate in the main relative to the orig. To replicate the orig, restore `scalefactor = 2`:

```
carryover_effect = case_when(
  treatment == 1 ~ 1,
  treatment == 0 & carryover_t_half == 0 ~ 0,
  treatment == 0 & carryover_t_half > 0 ~
    (1/2)^(2 * tsd / carryover_t_half),
  TRUE ~ 0
)
```

Note: `pm_functions.R` already has `scale_factor = 2` as default. The divergence is only in `simulation_clustered.R` which implements its own inline carryover computation.

## 10.4 Replication: Implement Ron Thomas correlation scaling (Critical)

The orig's `generateData.R` (lines 148-153) scales the biomarker-BR correlation at off-drug timepoints using a ratio of BR means:

```
correlations["bm", n1] <- ifelse(
  brtest[p],
  ifelse(brmeans[p] == 0, 0,
    (mm1 / mm0) * modelparam$c.bm),
  modelparam$c.bm)
```

where `brtest[p]` is TRUE when the raw (pre-carryover) BR mean was zero, `mm1` is the carryover-adjusted BR mean at the current timepoint, and `mm0` is the BR mean at the previous timepoint.

This scaling is the mechanism by which the biomarker- treatment interaction varies across on-drug and off-drug timepoints in the MVN covariance structure. The main's `simulation_clustered.R` does not implement this scaling – it uses a fixed `c.bm` for all BR-biomarker correlations regardless of treatment status. This is a fundamental methodological difference that must be resolved for faithful replication.

The interaction between this correlation scaling and the carryover parameters is the most likely source of the “power falls too rapidly” problem (see Section 10.5).

## 10.5 Diagnosis: Correlation matrix perturbation analysis (High)

Once the main replicates the orig's variable chain, the power dropoff can be diagnosed by examining how the biomarker-BR correlation varies across carryover levels.

With the orig's aggressive decay (`scalefactor = 2`, `t_half = 0.2`, `tsd >= 1` at first off-drug timepoint):

```
Carryover retention at week 11: 0.001
Correlation scaling: mm1/mm0 ≈ 0.001
Effective c.bm at week 11: 0.001 * 0.3 ≈ 0.0003
```

This near-zero correlation at off-drug timepoints means the off-drug observations contribute almost no information about the biomarker-treatment interaction. The result is effectively the same as having no off-drug data at all.

But with zero carryover ( $t_{\text{half}} = 0$ ), the off-drug BR mean is exactly zero and the correlation is set to zero. The off-drug observations also contribute no interaction information.

So the question becomes: **why does power differ between  $t_{\text{half}} = 0$  and  $t_{\text{half}} = 0.2$  if both produce near-zero off-drug correlations?**

Possible mechanisms to investigate:

1. **PD enforcement distortion.** The `make.positive.definite(sigma, tol = 1e-3)` step may redistribute variance differently when small nonzero correlations are present ( $\text{carryover} > 0$ ) versus exactly zero ( $\text{carryover} = 0$ ). Even small perturbations to the on-drug block of the covariance matrix could degrade the detectable interaction.
2. **BR mean contamination.** Even 0.1% residual BR at off-drug timepoints changes the Gompertz trajectory through the accumulation logic. At week 12 of the Hybrid design, `brmeans[6] = brmeans[5] * 0.001`, but `brmeans[5]` itself was `brmeans[4] * 0.001`. These cascading small values may interact with the correlation scaling in unexpected ways (e.g.,  $\text{mm1}/\text{mm0} = 0.001/0.001$  producing unstable ratios).
3. **Analysis model sensitivity.** The analysis model uses `Dbc` (the continuous treatment indicator), which in the orig incorporates the same carryover formula. When `Dbc` has small nonzero values off-drug instead of exactly zero, the `bm * Dbc` interaction has slightly different properties. The regression coefficient may be biased by the near-zero off-drug `Dbc` values that still carry some variance.

Recommended diagnostic script:

```
# For each carryover level (0, 0.1, 0.2):
# 1. Print the full 26x26 correlation matrix
# 2. Print eigenvalues before and after PD enforcement
# 3. Print the biomarker-BR correlations at each TP
# 4. Print the BR means at each TP per path
# 5. Compare sigma matrices element-wise between
#    t_half = 0 and t_half > 0
```

## 10.6 Diagnosis: Targeted power comparison (High)

Run a small simulation (50-100 iterations) with the replicated orig logic at three carryover levels and compare power curves. If the main's replicated results match the orig's published power curves, the replication is confirmed and the power dropoff is inherent to the orig's methodology. If they still differ, there is a remaining implementation gap to close.

## 10.7 Supporting changes

#	Action	Priority
7	Validation script: print all 5 vars for each design/path against hand-	High

#	Action	Priority
	computed reference values	
8	Unit tests for tsd, tod, carryover at boundary conditions	Medium
9	Document all convention choices (tsd anchor, tod reset, scale factor, correlation scaling) in a single reference document	Medium
10	Trim DESCRIPTION: remove unused database drivers, palmerpenguins, etc.	Low

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